# Daily Progress Report

Air–Sea Heat Flux Drivers and Trends (1993–2023)

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## 1 Objective

The objective of this work is to investigate long-term changes in the atmospheric and oceanic variables controlling air–sea heat exchange and to evaluate how classical bulk aerodynamic physics compares with observed turbulent heat fluxes from ERA5 reanalysis data.

This analysis is intended as a **physical grounding step** prior to the development of a Physics-Informed Neural Network (PINN) model, ensuring that the dominant physical drivers and their long-term behavior are well understood.

**Study Region:** Longitude 40°E–120°E, Latitude 60°S–30°N

**Study Period:** 1993–2023

## 2 Data Sources and Variables

ERA5 reanalysis data accessed via Google Earth Engine (GEE) was used.

### 2.1 Atmospheric Variables (Monthly)

- 10 m wind components ($u_{10}, v_{10}$)
- 2 m air temperature ($T_{2m}$)
- 2 m dewpoint temperature ($T_d$)
- Surface pressure ($p_s$)

### 2.2 Oceanic and Flux Variables (Hourly, Aggregated to Monthly)

- Sea surface temperature (SST)
- Surface latent heat flux (LHF)
- Surface sensible heat flux (SHF)

Hourly data were aggregated to monthly means to maintain consistency with the long-term trend analysis.

# 3 Derived Physical Quantities

Using the primary ERA5 variables, the following physically meaningful quantities were derived:

- Wind speed: $U = \sqrt{u_{10}^2 + v_{10}^2}$
- Air density: $\rho = \frac{p_s}{RT_{2m}}$
- Air–sea temperature difference: $\Delta T = T_s - T_a$
- Specific humidity at air and sea surface
- Air–sea humidity difference: $\Delta q = q_s - q_a$

These variables form the basis of bulk aerodynamic heat flux formulations.

# 4 Bulk Flux Formulation

Classical bulk flux estimates were computed as a physics baseline.

## 4.1 Sensible Heat Flux

$$Q_H = \rho C_p C_H U (T_s - T_a)$$

## 4.2 Latent Heat Flux

$$Q_L = \rho L_v C_E U (q_s - q_a)$$

where $C_H$ and $C_E$ are bulk transfer coefficients.
These estimates serve as a reference against which observed ERA5 fluxes are compared.

# 5 Long-Term Trends of Flux Drivers

## 5.1 Sea Surface Temperature (SST)

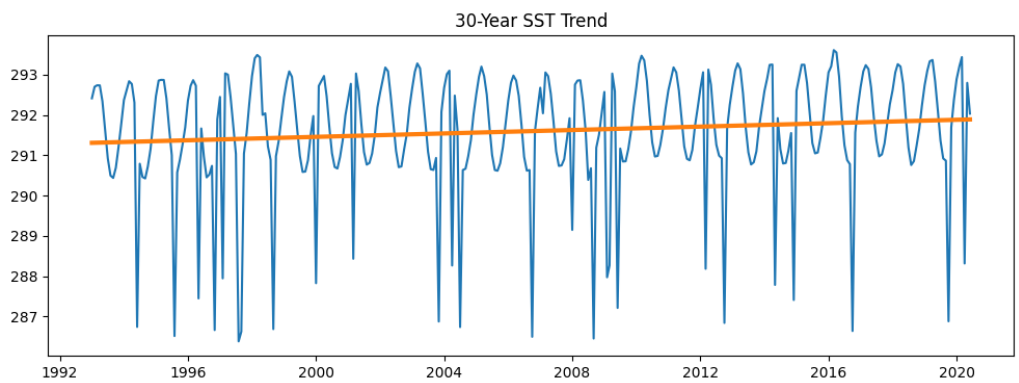


Figure 1: Regional mean sea surface temperature (SST) trend from 1993 to 2023.

SST exhibits a persistent warming trend across the study period. This warming represents the dominant long-term signal influencing air–sea heat exchange.

**Physical implication:** Higher SST increases evaporation potential and atmospheric moisture capacity.

### 5.2 Near-Surface Air Temperature

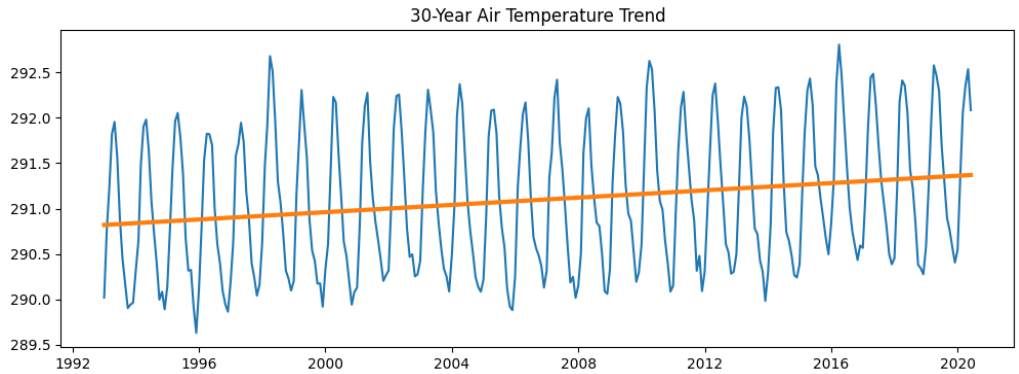


Figure 2: Near-surface air temperature trend (2 m) over the study region.

Air temperature increases at a rate comparable to SST, indicating coupled warming of the ocean and atmosphere.

**Implication:** The air–sea temperature difference remains relatively stable, limiting strong long-term changes in sensible heat flux.

### 5.3 Wind Speed

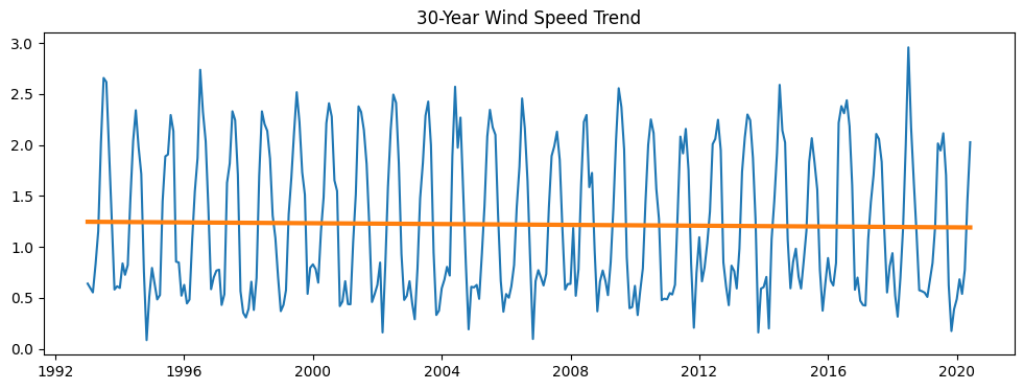


Figure 3: Mean near-surface wind speed trend from 1993 to 2023.

Wind speed exhibits comparatively weak long-term trends, though it remains a dominant driver of short-term variability.

### 5.4 Humidity and Moisture Gradient

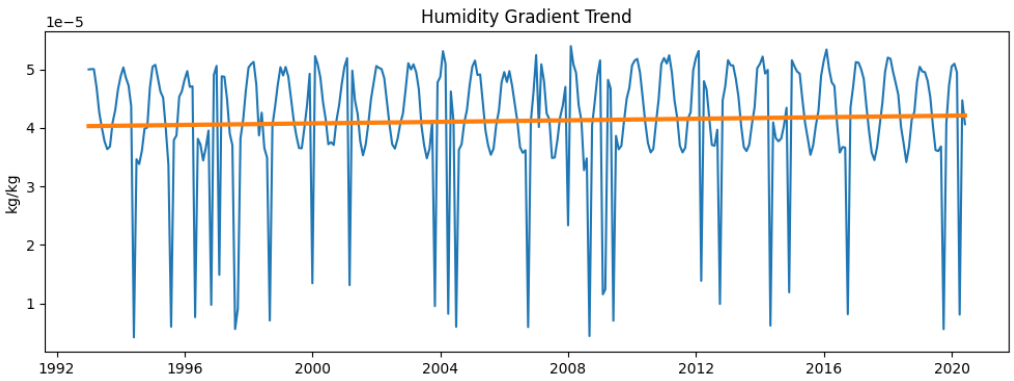


Figure 4: Air–sea humidity difference showing an increasing long-term tendency.

Increasing atmospheric moisture and humidity gradients are observed, supporting enhanced evaporation over time.

## 6 Bulk Flux Trends vs Observed ERA5 Fluxes

### 6.1 Latent Heat Flux

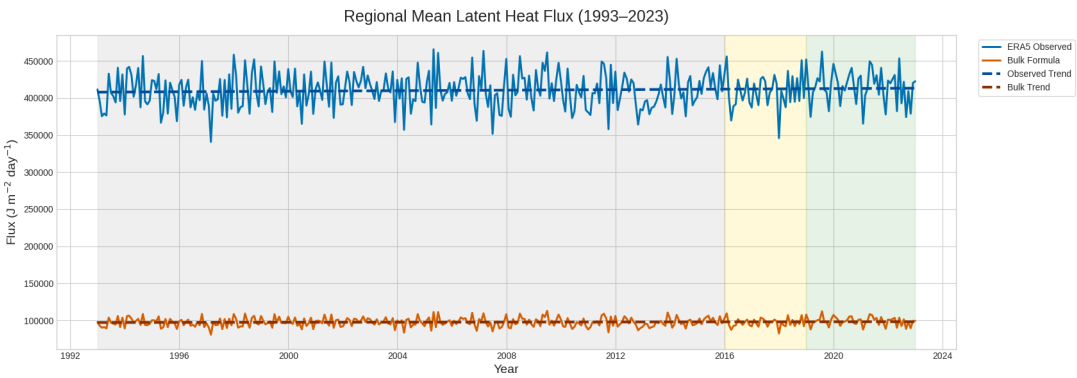


Figure 5: Comparison of bulk-estimated and ERA5 latent heat flux trends.

Both bulk-estimated and observed latent heat flux show increasing trends, indicating that bulk physics captures the dominant first-order behavior.

However, observed fluxes contain additional variability not represented in bulk formulations.

### 6.2 Sensible Heat Flux

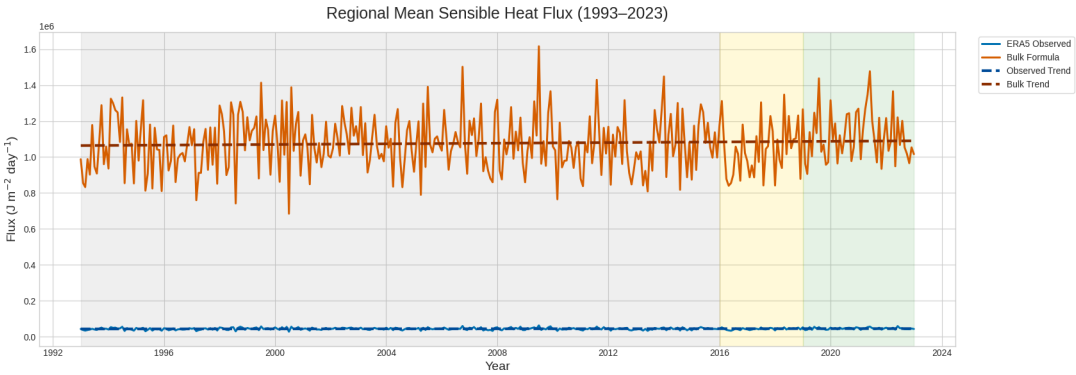


Figure 6: Comparison of bulk-estimated and ERA5 sensible heat flux trends.

Sensible heat flux trends are weaker and more variable, reflecting the relatively stable air–sea temperature difference.

## 7 Driver–Flux Relationships (Qualitative)

At this stage, **formal statistical correlations have not been computed**. The analysis focuses on qualitative consistency with physical theory.

- Latent heat flux responds primarily to SST, wind speed, and humidity gradient.
- Sensible heat flux is controlled mainly by air–sea temperature difference.

These behaviors align with bulk aerodynamic theory, motivating future quantitative analysis.

## 8 Scientific Implications

- Evidence supports an intensifying hydrological cycle driven by ocean warming.
- Bulk physics captures first-order trends but not full variability.
- Missing processes likely include boundary-layer stability and mesoscale dynamics.

## 9 Role in PINN Development

This study establishes a physical baseline for PINN development.

The intended PINN approach will:

- Preserve known physical relationships
- Learn unresolved atmospheric processes
- Improve flux estimation beyond classical bulk models

## 10 Summary of Daily Progress (12/02/2026)

- Implemented training strategy for 30 years of historical data using random batch sampling.
- Avoided full data download by sampling directly from GEE within Colab.
- Established physically consistent trend analysis framework.
- Initiated planning for spatial maps and visualizations.

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